A5 Implementation Report: Carbon Offset Investment Dashboard

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Motivation

I implemented my A4 "Dashboard A" for carbon offset managers who need to choose cost-effective projects (e.g., reforestation or carbon capture) across countries while considering risk. The dashboard supports:

- **T1**: Find suitable regions for carbon offset projects.
- **T2**: Compare cost-effectiveness across regions (I use climate hazard/risk as a proxy dimension because no explicit cost data exists in the source).
- **T3**: Identify areas losing forests but with regrowth potential.

I chose this design because it combines a geographic overview with multi-criteria filtering and trend context, which matches the decision flow of this user group. I focused on clear labels, units, and consistent color use to address the A4 feedback.

Prototyping / Design process

- **Data prep**: With `prep_data.py` I reshaped the raw CSVs (`13_Forest_and_Carbon.csv`, `15_Climate-driven_INFORM_Risk.csv`, `03_National_Greenhouse_Gas_Emissions_...csv`) into smaller JSONs:
  - `data/country_summary.json` (carbon stock, forest change %, hazard, risk, vulnerability, coping capacity).
  - `data/emissions_timeseries.json` (total GHG including land-use, Gas Type = "Greenhouse gas").

  Countries are matched by ISO3; missing values stay empty rather than zero-filled. The map uses `data/world-110m.json`.
- **Changes driven by A4 feedback**: Added titles, axis labels/units, legends (color and size), tooltips with data years; multiple selection methods (map/scatter click, chips, search) instead of map-only; single-purpose colors; explicit "latest data years" note; one-screen layout with brush filters (no tabs, dropdowns, or sliders).
- **Design adjustments**: Light, high-contrast theme; white strokes on scatter points; gray "No data" on the map; risk gradient + forest-area size legend on scatter; selection capped at five with chips for clarity; parallel axes focused on the most relevant fields (carbon, forest change %, hazard, risk, vulnerability, coping).

- **What I corrected during prototyping**: Reduced chart heights to avoid overlap; clarified missing-data state on the map; simplified legends so they are readable at a glance.

Using the prototype, I can quickly find countries with high carbon stocks but lower hazard scores, and I can compare forest change against emissions to identify places with loss but strong carbon potential. The scatter plot reveals "sweet spot" candidates, while the parallel coordinates highlight outliers across multiple criteria.

One idea that did not work on the real data was a denser parallel-coordinates design with more axes; it became too cluttered and slowed interactions. I also tested stronger color saturation, but it reduced readability, so I switched to a cleaner, high-contrast palette.

Implementation details

- **Tech stack**: Plain D3 v7 plus topojson. Static files: `index.html`, `style.css`, `main.js`, data JSONs, `data/world-110m.json`. No frameworks.
- **Views and interactions**:
  - **Carbon Stock Map** (top-left in Fig. 1): Sequential green fill for latest carbon stock (Mt); gray = no data. Hover shows value and year; click toggles selection; search highlights matches.
  - **Risk vs Carbon Scatter** (top-right in Fig. 1): X = hazard (0–10), Y = carbon (log), size = forest area, color = risk. Legends for risk gradient and size; hover shows values and year; click to select.
  - **Parallel Coordinates** (bottom-left in Fig. 1): Axes for carbon (log), forest change %, hazard, risk, vulnerability, coping. Brush any axis to filter; a badge shows how many filters are active; filters apply to all views.
  - **Emissions Timeline** (bottom-right in Fig. 1): Lines for up to five selected countries, using total GHG including land-use. Gridlines for readability; hover shows the last value and year.
  - **Selection UX**: Chips under the header with an "x" per country, plus a clear-all button; search input highlights matching countries; selection propagates across all views.
- **Data coverage**: About 200 countries with carbon/risk; 213 with emissions series. Latest carbon/risk values are mostly 2022–2023; emissions span varies by country (up to 1970–2023 where available).
- **Deployment**: Runs as a static page. Launch with a simple server (e.g., `python3 -m http.server 8000`) and open `index.html`.

The implementation addresses the tasks as follows: T1 is supported by the map plus filtering in parallel coordinates; T2 is supported by the scatter plot comparison of hazard vs carbon; T3 is supported by forest change in parallel coordinates together with the emissions timeline for context.

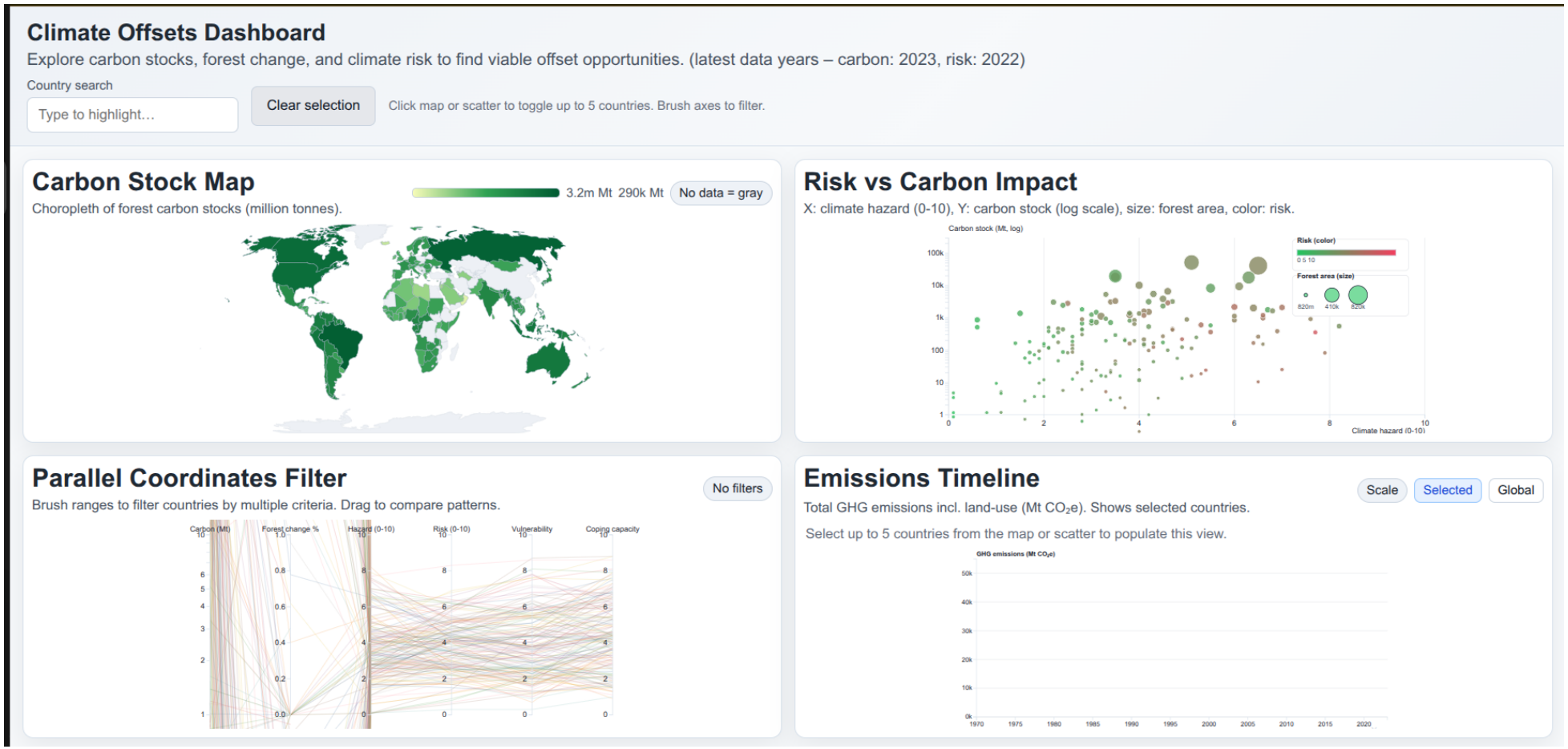

Figure 1: Full-page screenshot of the implemented dashboard.

Discussion

- **Task fit**:
  - T1: Map for geographic overview + parallel brushes to narrow candidates.
  - T2: Scatter shows hazard (as cost/risk proxy) vs carbon impact with size for forest area.
  - T3: Forest change % in parallel coords plus carbon and emissions trends highlight loss vs potential.
- **Usability**: All views are visible on one screen; legends and labels are explicit; "no data" is clearly marked; multiple ways to select/deselect keep the flow smooth.
- **Performance**: Limiting selection to five keeps timelines legible; filtering reduces line clutter. Data is pre-aggregated to compact JSONs for fast load.
- **Gaps / next steps**: No real cost data (hazard/risk used as proxy); parallel coordinates still needs a brief user hint; keyboard navigation and mobile layout could improve accessibility.

I learned that the design needed more explicit labels and legends once real data was shown, and that limiting interactions (like the maximum selected countries) helps keep comparisons readable.

Conclusion

I translated the A4 design into a working D3 dashboard with linked views, clear labeling, and consistent encodings. A4 feedback on detail, selection options, and color semantics is addressed. The dashboard supports T1–T3 and remains usable on a single screen.

AI usage

I used DeepSeek to brainstorm code structure and check D3 patterns.